

Source Table | Source Column | Source Type | Target Table | Target Column | Target Type | Transform Rule | Validation Rule | Business Logic | Comments

customer | CustId | | customer | CustId | | Direct mapping | Not null, unique | Primary key for customer identification |

customer | Name | | customer | Name | | Direct mapping | Not null | Stores customer name |

customer | EmailId | | customer | EmailId | | Direct mapping | Format must be valid email, not null | Unique contact detail |

customer | Region | | customer | Region | | Direct mapping | | Customer's region |

order | OrderId | | order | OrderId | | Direct mapping | Not null, unique | Primary key for order identification |

order | ItemName | | order | ItemName | | Direct mapping | Not null | Item description |

order | PricePerUnit | | order | PricePerUnit | | Direct mapping | Positive value, not null | Price per individual item |

order | Qty | | order | Qty | | Direct mapping | Positive integer, not null | Quantity of items ordered |

order | Date | | order | Date | | Direct mapping | Date format (YYYY-MM-DD), not null | Order date |

order | CustId | | order | CustId | | Direct mapping | Must exist in customer.CustId | Foreign key to customer |

order | | TotalAmount | order | TotalAmount | | PricePerUnit \* Qty | Must be  $\geq 0$  | Aggregated column for amount calculation | New column derived from existing fields

ordersummary | CustId | | ordersummary | CustId | | From joined customer|order on CustId | Not null | Preserved for SCD2 processing |

ordersummary | Name | | ordersummary | Name | | From customer | Not null | Preserved for SCD2 processing |

ordersummary | EmailId | | ordersummary | EmailId | | From customer | Email format, not null | Preserved for SCD2 processing |

ordersummary | Region | | ordersummary | Region | | From customer | | Preserved for SCD2 processing |

ordersummary | OrderId | | ordersummary | OrderId | | From order | Not null | Preserved for SCD2 processing |

ordersummary | ItemName | | ordersummary | ItemName | | From order | Not null | Preserved for SCD2 processing |

ordersummary | PricePerUnit | | ordersummary | PricePerUnit | | From order | Positive value, not null | Preserved for SCD2 processing |

ordersummary | Qty | | ordersummary | Qty | | From order | Positive integer, not null | Preserved for SCD2 processing |

ordersummary | Date | | ordersummary | Date | | From order | Date format (YYYY-MM-DD), not null | Preserved for SCD2 processing |

ordersummary | | StartDate | ordersummary | StartDate | | Set on record creation | Must be valid date | SCD2 effective start date |

ordersummary | | EndDate | ordersummary | EndDate | | Updated on customer change | Can be null for active, date for inactive | SCD2 effective end date |

ordersummary | | ActiveFlag | ordersummary | ActiveFlag | | Set to 'Y' for current, 'N' for inactive | 'Y' or 'N' | Indicates active/inactive status for SCD2 |

customeraggregatespend | Name | | customeraggregatespend | Name | | Group by from ordersummary | Not null | Aggregated field for spend calculation |

customeraggregatespend | TotalAmount | | customeraggregatespend | TotalAmount | | SUM(TotalAmount) grouped by Name, Date | TotalAmount ≥ 0 | Aggregated spend |

customeraggregatespend | Date | | customeraggregatespend | Date | | Group by from ordersummary | Date format (YYYY-MM-DD), not null | Aggregated spend per date |

Validation Rule: Remove all Null and duplicate records before loading data into target tables.

#### Business Logic Notes:

- Primary and foreign key constraints enforced throughout all transformations.
- Data quality and pattern validation on each field based on rules above.
- SCD2 logic for ordersummary: on customer change, set EndDate of previous records and insert/update with new customer data.

#### Comments:

- Datatypes and constraints to be defined based on implementation context ().
- Reference data for validation to be derived from customer and order tables as per transformation steps.
- All transformation and scheduling implemented via delta live tables as per requirements.

#### Load Frequency and Scheduling Notes:

- All steps and updates must be scheduled according to operational batch windows ().
- SCD2 and aggregation updates must run post-source refresh and data ingestion completion.
- Frequency and triggers for pipeline execution to be defined ().

#### Data Lineage Notes and Metadata:

- Full lineage tracked from original source CSV, through all tables and transformation stages, to final aggregated outputs.